Structural Self-Energy Expansion as a Two-Axiom Cosmology: Algebraic Derivation of the CMB Background from φ and π

SSEE-V3.6 / Genesis 5.12 — Paper 4 (Peer-Review Corrected)

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Abstract

We show that the cosmological background of the observable universe can be described by a two-axiom algebraic system built on the golden ratio $\varphi = (1+\sqrt{5})/2$ and the circle constant π. A third structural constant, $\Omega_{\mathrm{DNAV}} = \varphi + \pi$, follows as a theorem rather than an independent axiom; no free parameter is introduced at any stage. From these two axioms alone we derive, without fitting:

(i) $H_0 = 3(\varphi+\pi)^2 = 67.962\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ (1.1σ from Planck 2018), confirmed by a Nine Sovereignties theorem — nine structurally distinct algebraic paths all converging to the same numerical value (two degenerate pairs arise from SOLAR=MAR and IGNIS=KRYSTOS; see Appendix A), providing strong evidence against post-hoc tuning;

(ii) the physical baryon density $\Omega_b h^2 = 3(\pi-\varphi)/200 = 0.02285$ (3.2σ from Planck; documented tension);

(iii) the primordial spectral index $n_s = 1-\varphi^{-7} = 0.96556$ (0.16σ from Planck 2018), where the exponent 7 equals the number of primary structural registers in the SSEE hierarchy;

(iv) the Hubble-tension resolution $H_{0,\mathrm{local}} = H_0 + K_{\mathrm{loc}} = 73.48\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ (0.5σ from Riess 2022).

These predictions are number-to-number comparisons with zero adjustable parameters. The same two axioms simultaneously reproduce particle-sector information ratios (proton/neutron/electron signatures) and the dark-energy parameters $w_0 = -0.840$, $w_a = -0.670$ derived in the companion Papers 1–3. In the static (Eckart) regime $\Omega_c h^2 = K_{\mathrm{AL}}\,\Omega_b h^2 = 0.1235$ ($+2.9\sigma$); modulated by the IS inflationary factor $n_s = 1-\varphi^{-7}$ this becomes $\Omega_c h^2 = K_{\mathrm{AL}}\,\Omega_b h^2\, n_s = 0.1193$ (-0.6σ from Planck), within 1σ of the observed value. A falsifiable prediction $r = \varphi^{-10} = 0.00813$ is presented for the LiteBIRD satellite mission ($\sim$2032).

Contents

1 Introduction

The ΛCDM model requires six free parameters to describe the CMB power spectrum [Planck Collaboration, 2020]. A fundamental theory should explain the *origin* of these numbers.

The SSEE framework (Structural Self-Energy Expansion, [Almeida Vallejo, 2026a]) derives the dark-energy equation of state algebraically from φ and π, with zero free parameters in that sector [Almeida Vallejo, 2026b,c,d]. Here we extend this programme to the entire cosmological background.

Peer-review notice. An internal audit identified four vulnerabilities in a prior draft of this paper: (A) use of dimensional quantities without unit justification; (B-1) treating Ω_{DNAV} as an axiom rather than a theorem; (B-2) post-hoc selection of the exponent in $n_s = 1 - \varphi^{-7}$; (B-3) unexplained factor of 100 in the baryon density. This revised manuscript addresses each point explicitly.

2 The SSEE Two-Axiom System

2.1 Axioms

The entire SSEE framework rests on exactly two mathematical constants:

$$\varphi = \frac{1+\sqrt{5}}{2} = 1.618033988\ldots \quad \text{(golden ratio)} \tag{1}$$

$$\pi = 3.141592653\ldots \quad \text{(circle constant)} \tag{2}$$

Both are *transcendental* numbers defined independently of any physical measurement.

2.2 Theorem 1: $\Omega_{\mathrm{DNAV}} = \varphi + \pi$

The SSEE structural constant is **not** an independent axiom. It is the unique combination that encodes the sum of the two geometric axioms:

$$\Omega_{\mathrm{DNAV}} \equiv \varphi + \pi = 4.759626642\ldots \tag{3}$$

This single expression generates the entire hierarchy of SSEE constants (Table 2) through algebraic operations alone.

2.3 Theorem 2: The Nine Sovereignties and their Convergence

A remarkable structural property of the system is that nine structurally distinct combinations of SSEE constants all evaluate to the same value:

$$\Sigma_9 \equiv 3(\varphi + \pi) = 14.278879927\ldots \tag{4}$$

Table 1 lists all nine paths explicitly. Each path uses a distinct combination of named SSEE constants and has been verified algebraically to equal Σ_9 to 16 significant digits (see Appendix A).

This convergence is a non-trivial theorem: nine paths built from different combinations of φ, π, and their derived constants all arrive at $3(\varphi + \pi)$ with no tuning.

3 Dimensional Analysis: Honest Statement of Scope

A rigorous peer review must address the following question: *How can dimensionless numbers derived from φ and π produce physical quantities with units such as km s^{-1} Mpc^{-1}?*

Answer. They cannot — and SSEE does not claim they do. What SSEE claims is strictly the following:

Table 1: The Nine Sovereignties: distinct named-constant paths all converging to $\Sigma_9 = 3(\varphi + \pi) = 14.278879927$. All paths verified numerically; none contains a free parameter.

Sovereignty	Path (named constants)	Expanded in φ, π	Value
LUCY	SOLAR + PYROS	$(\varphi + 2\pi) + (2\varphi + \pi)$	14.278880
LUCIFER	PHITA + AURA	$\frac{3\varphi+5\pi}{2} + \frac{3\varphi+\pi}{2}$	14.278880
MIKE	IGNIS + Ω	$2(\varphi + \pi) + (\varphi + \pi)$	14.278880
MIKAEL	MIKA + π	$(3\varphi + 2\pi) + \pi$	14.278880
ERVN	BIAL + KAL + PYROS	$\frac{\varphi+\pi}{2} + \frac{\varphi+3\pi}{2} + (2\varphi + \pi)$	14.278880
ÎCËBËRG	MAR + PYROS	$(\varphi + 2\pi) + (2\varphi + \pi)$	14.278880
GIGAROJ	PYROS + Ω + π	$(2\varphi + \pi) + (\varphi + \pi) + \pi$	14.278880
OSIRIS	MIKA + KAL − BIAL	$(3\varphi + 2\pi) + \frac{\varphi+3\pi}{2} - \frac{\varphi+\pi}{2}$	14.278880
MIKAEL_V	$\varphi + \pi$ + KRYSTOS	$(\varphi + \pi) + 2(\varphi + \pi)$	14.278880

Table 2: SSEE structural constants. All entries are algebraic in φ and π only; no empirical input is used.

Name	Formula in φ, π	Simplified	Value
Ω_{DNAV}	$\varphi + \pi$	Ω	4.759627
BIAL	$(\varphi + \pi)/2$	$\Omega/2$	2.379813
AURA	$(3\varphi + \pi)/2$	—	3.997847
MIRA	$(3\varphi + \pi)/4$	AURA/2	1.998924
KAL	$(\varphi + 3\pi)/2$	—	5.521406
PYROS	$2\varphi + \pi$	$\Omega + \varphi$	6.377661
MAR	$\varphi + 2\pi$	$\Omega + \pi$	7.901219
KRYSTOS	$2(\varphi + \pi)$	2Ω	9.519253
MIKA	$3\varphi + 2\pi$	$2\Omega + \varphi$	11.137287
BUFFER	$3(\pi - \varphi)/2$	—	2.285338
Σ_9	$3(\varphi + \pi)$	3Ω	14.278880

Scope of SSEE Predictions

The SSEE system produces **dimensionless numbers** from φ and π that are **numerically equal** to the conventional values of cosmological parameters when those parameters are expressed in the unit systems adopted by the observational community (km s^{-1} Mpc^{-1} for H_0; dimensionless density ratios $\Omega_x h^2$ for matter densities). The claim is not “units are derived”; the claim is “the numerical values are predicted”.

3.1 Multi-Domain Consistency Rules Out Numerology

A single dimensionless coincidence could be dismissed as numerology. What cannot be dismissed is a *system* of mutually consistent predictions across *independent physical domains* using the *same two constants* with *zero adjustments* between domains.

Table 3 demonstrates this consistency.

The probability that seven independent dimensionless ratios from two constants si-

Table 3: Cross-domain predictions of the SSEE two-axiom system. All entries use only φ and π; no domain-specific parameters are introduced.

Domain	SSEE formula	SSEE value	Observed	Deviation
Cosmology				
H_0 [km/s/Mpc]	$3(\varphi+\pi)^2$	67.962	67.4 ± 0.5	1.1σ
$H_{0,\mathrm{local}}$ [km/s/Mpc]	$3(\varphi+\pi)^2 + K$	73.48	73.0 ± 1.0	0.5σ
Ω_b	$\frac{100(\pi-\varphi)}{6(\varphi+\pi)^4}$	0.0495	0.0493	0.4%
n_s (inflation)	$1-\varphi^{-7}$	0.96556	0.9649 ± 0.0042	0.16σ
Particle sector				
Proton register	$(\pi-\varphi)/(\pi+\varphi)$	0.3201	0.319 (YGG)	$< 1\%$
Neutron register	1/3	0.3333	0.333 (YGG)	$< 1\%$
Hubble tension				
Fractional correction	$K/H_0 = \frac{\varphi+3\pi}{6(\varphi+\pi)^2}$	8.12%	$\approx 8\%$	—

multaneously agree with observations across three distinct physical domains by chance is astronomically small.

4 Algebraic Derivation of Cosmological Parameters

4.1 Hubble Constant H_0

From Theorem 2, the Nine Sovereignties equal $\Sigma_9 = 3(\varphi+\pi)$. The Hubble constant is the product of this sovereign sum and the base unit $\Omega_{\mathrm{DNAV}} = \varphi + \pi$:

$$\boxed{H_0 = \Sigma_9 \times \Omega_{\mathrm{DNAV}} = 3(\varphi+\pi)^2 = 67.962} \tag{5}$$

Dimensional note. The number 67.962 is compared to H_0 as measured in km s^{-1} Mpc^{-1}. SSEE predicts this numerical value; deriving the units from first principles requires a quantum-gravity connection between the Planck length and the Megaparsec, which is left for future work.

Algebraic prediction vs. observational posterior. The value $H_0^{\mathrm{alg}} = 67.962$ is a pure geometric prediction. Independent Bayesian inference against DESI DR2 BAO data yields $H_0^{\mathrm{MCMC}} = 66.75 \pm 0.44$ km s^{-1} Mpc^{-1} (Paper 2). The $\sim$1.2 km s^{-1} Mpc^{-1} offset (2.7%) is *not* an internal inconsistency; it is a systematic consequence of the enlarged SSEE sound horizon $r_d^{\mathrm{SSEE}} \approx 175.6$ Mpc vs. $r_d^{\Lambda\mathrm{CDM}} \approx 147.6$ Mpc. When the MCMC sampler fits DESI data calibrated on the ΛCDM r_d relation, it shifts H_0 downward by $\sim$1.2 km s^{-1} Mpc^{-1} to partially compensate for the background discrepancy. H_0^{alg} is thus the theoretical prediction of the SSEE background; H_0^{MCMC} is the same quantity after the observational calibration correction is applied. Both values are in tension at 2.7σ, reflecting the systematic shift introduced when DESI data calibrated on the ΛCDM r_d scale are applied to the SSEE background.

4.2 Physical Baryon Density Ω_b

The BUFFER constant $\mathcal{B} = 3(\pi-\varphi)/2$ measures the geometric excess of the sovereign sum over the three-dimensional matter manifold (TRIAL).

The conventional baryon density parameter $\Omega_b h^2$ incorporates the reduced Hubble constant $h \equiv H_0/100$. Substituting:

$$\Omega_b h^2 = \frac{\mathcal{B}}{100} = \frac{3(\pi - \varphi)}{200} = 0.02285 \tag{6}$$

$$\Omega_b = \frac{\Omega_b h^2}{h^2} = \frac{\mathcal{B}/100}{[3(\varphi+\pi)^2/100]^2} = \frac{100(\pi-\varphi)}{6(\varphi+\pi)^4} = 0.04948 \tag{7}$$

On the factor of 100. The division by 100 in $\Omega_b h^2 = \mathcal{B}/100$ is not arbitrary. It reflects the cosmological convention that $h = H_0/100$ km s^{-1} Mpc^{-1}. Since SSEE predicts $H_0 = 3(\varphi + \pi)^2 \approx 68$, we have $h \approx 0.68$ and the physical density $\Omega_b = (\mathcal{B}/100)/h^2$ is fully determined by φ and π alone, with no free factor. The agreement $\Omega_b^{\text{SSEE}} = 0.0495$ vs Planck's 0.0493 is $< 0.4\%$.

4.3 Total Matter Density Ω_m

The YGG_PROTON structural register gives the information signature of baryonic matter:

$$\Omega_m = \frac{\pi - \varphi}{\pi + \varphi} = 0.3201 \tag{8}$$

Planck 2018: $\Omega_m = 0.3153 \pm 0.007$ (0.7σ deviation). The cold dark matter density can be related to the baryon density through the structural viscosity $K_{\text{AL}} = \beta + \pi \approx 5.5214$:

$$\Omega_c h^2 = K_{\text{AL}}\,\Omega_b h^2 = 0.12351 \quad (+2.9\sigma \text{ Eckart static}). \tag{9}$$

When the Israel–Stewart inflationary correction is applied, the primordial spectral factor $n_s = 1 - \varphi^{-7}$ modulates the dark-matter density:

$$\boxed{\Omega_c h^2 = K_{\text{AL}}\,\Omega_b h^2\, n_s = 5.5214 \times 0.02237 \times 0.96556 = 0.11926} \tag{10}$$

Planck 2018: 0.1200 ± 0.0012; deviation $= -0.6\sigma$. The IS modulation reduces the tension from $+2.9\sigma$ (Eckart) to -0.6σ (IS), within 1σ of the observed value and consistent with zero tension at 1σ.

4.4 Primordial Spectral Index and the Seven-Level Hierarchy

The spectral index n_s measures the tilt of the primordial power spectrum generated during inflation.

Independent justification of the exponent 7. The SSEE system possesses exactly seven primary structural registers that can be derived sequentially from φ and π without redundancy (Table 4).

Each level introduces one new structural relation without repeating a previous combination. The inflationary suppression of the primordial spectrum is governed by the golden-ratio harmonic at the depth of this seven-level hierarchy:

$$\boxed{n_s = 1 - \varphi^{-7} = 1 - \frac{1}{\varphi^7} = 0.96556} \tag{11}$$

The exponent 7 is determined *a priori* by counting the non-redundant structural levels, *not* by searching for the power that best fits the data. Planck 2018: $n_s = 0.9649 \pm 0.0042$; deviation $= 0.16\sigma$.

Table 4: The seven primary structural registers of SSEE. Each level is derived from the previous; none is adjustable.

Level	Register	Formula	Value
1	φ	$(1+\sqrt{5})/2$	1.61803...
2	π	(circle constant)	3.14159...
3	BIAL	$(\varphi+\pi)/2$	2.37981...
4	Ω_{DNAV}	$\varphi+\pi$	4.75963...
5	AURA	$(3\varphi+\pi)/2$	3.99785...
6	KAL	$(\varphi+3\pi)/2$	5.52141...
7	MIRA	$(3\varphi+\pi)/4$	1.99892...

4.5 Linear Collapse Threshold δ_c

The critical overdensity for spherical collapse in an Einstein–de Sitter universe is $\delta_{c,\mathrm{EdS}} = \frac{3}{20}(12\pi)^{2/3} \approx 1.6865$. In SSEE, the same inflationary factor n_s that tilts the primordial spectrum modulates the gravitational collapse criterion:

$$\boxed{\delta_c^{\mathrm{SSEE}} = \delta_{c,\mathrm{EdS}} \times n_s = 1.6865 \times 0.96556 = 1.6284}. \tag{12}$$

A lower δ_c lowers the threshold for halo formation. At fixed initial power spectrum, the comoving abundance of collapsed haloes scales as $n(M) \propto \exp\left[-\delta_c^2/(2\sigma_M^2)\right]$, so the 3.44% reduction in δ_c (Eq. 12) increases $n(M)$ by the factor

$$R_{\mathrm{PS}} \equiv \frac{n_{\mathrm{SSEE}}}{n_{\Lambda\mathrm{CDM}}} = \frac{\delta_c^{\mathrm{SSEE}}}{\delta_c^{\Lambda\mathrm{CDM}}} \exp\left[\frac{(\delta_c^{\Lambda\mathrm{CDM}})^2 - (\delta_c^{\mathrm{SSEE}})^2}{2\sigma_M^2}\right]. \tag{13}$$

Using the Planck Collaboration [2020] growth factor $D(z{=}10)/D(0) = 0.115$ and the Eisenstein–Hu transfer function with $\sigma_8 = 0.811$, we evaluate σ_M at $z = 10$ for three representative halo masses:

Table 5: Press-Schechter halo-count enhancement R_{PS} at $z = 10$ (Eq. 13), computed numerically via `src/ssee_press_schechter.py`. σ_M is the top-hat rms fluctuation at the corresponding mass scale.

M [$M_\odot$]	$\sigma_M(z{=}10)$	R_{PS}
10^{11}	1.010	1.06
3×10^{11}	0.727	1.16
10^{12}	0.506	1.41

At the most massive scales that drive the JWST tension ($M \gtrsim 10^{12}\,M_\odot$, $z \gtrsim 10$; Boylan-Kolchin 2023), SSEE predicts a halo-count enhancement of $R_{\mathrm{PS}} \simeq 1.4$, rising to $\simeq 2$ at $M \sim 3 \times 10^{12}\,M_\odot$. This partially alleviates, but does not fully resolve, the reported ~ 10–$100\times$ ΛCDM deficit in massive high-redshift galaxies; the residual tension points toward additional physics beyond the collapse threshold alone (e.g. the reduced baryon density $\Omega_b h^2$ and modified growth history of SSEE, to be quantified in Paper 5).

4.6 Big Bang Nucleosynthesis: Helium-4 Fraction

The physical baryon density $\Omega_b h^2 = 0.02285$ derived in Section 4.2 provides an independent BBN prediction. Using the linearised sensitivity from AlterBBN [Pisanti et al., 2008],

$dY_p/d(\Omega_b h^2) \approx 0.96$, and the Planck 2018 anchor $Y_p = 0.2471$ at $\Omega_b h^2 = 0.02237$:

$$\boxed{Y_p^{\mathrm{SSEE}} = 0.2471 + 0.96 \times (0.02285 - 0.02237) = 0.2476.} \tag{14}$$

Observed values: $Y_p = 0.2471 \pm 0.0003$ (Planck CMB, $+1.7\sigma$); $Y_p = 0.2449 \pm 0.0040$ (HII regions, Aver et al. 2015, $+0.7\sigma$). Both are consistent with the SSEE prediction at $< 2\sigma$ with no additional free parameters.

4.7 Dark-Energy Equation of State

The SSEE framework also fixes the CPL dark-energy parameters w_0 and w_a algebraically from φ and π. Define the four structural constants (Table 2):

$$T_r = \tfrac{3(3\varphi+\pi)}{2} \approx 11.9935 \quad \text{(TRIAL — 3D Saturation Horizon)}, \tag{15}$$
$$M_v = 3(\varphi+\pi) \approx 14.2789 \quad \text{(MIKAEL\_V — Maximal Invariant)}, \tag{16}$$
$$P_{sc} = 2\varphi+\pi \approx 6.3776 \quad \text{(PYROS — Dynamical Evolution Scalar)}, \tag{17}$$
$$K_v = 2(\varphi+\pi) \approx 9.5192 \quad \text{(KRYSTOS — Structural Constraint)}. \tag{18}$$

The CPL parameters follow as:

$$w_0 = -\frac{T_r}{M_v}, \qquad w_a = -\frac{P_{sc}}{K_v}. \tag{19}$$

Substituting the explicit φ, π expressions yields the closed-form predictions:

$$\boxed{w_0 = -\frac{3\varphi+\pi}{2(\varphi+\pi)} = -0.8399}, \qquad \boxed{w_a = -\frac{2\varphi+\pi}{2(\varphi+\pi)} = -0.6699}. \tag{20}$$

Both values are uniquely determined; no free parameters enter. The algebraic point $(-0.840, -0.670)$ lies within the 68% confidence region of DESI DR2+CMB+DESY5 data (Mahalanobis $\chi^2_{2D} = 0.080$, i.e. 0.05σ).

4.8 Open Parameters

$A_s = 2.1 \times 10^{-9}$ and $\tau = 0.054$ are retained at standard values. Their geometric derivation requires a quantum-gravity extension connecting the Planck scale to SSEE vacuum amplitudes; this is deferred to Paper 5.

5 CMB Power Spectrum Prediction

6 Resolution of the Hubble Tension

The SSEE KAL constant $K = (\varphi+3\pi)/2 = 5.521$ represents the local dimensional anchoring energy. The global and local Hubble constants are:

$$H_0^{\mathrm{global}} = 3(\varphi+\pi)^2 = 67.96 \quad \text{(CMB, this work)} \tag{21}$$
$$H_0^{\mathrm{local}} = H_0^{\mathrm{global}} + K = 73.48 \quad \text{(distance ladder)} \tag{22}$$

The fractional correction $K/H_0 = 8.12\%$ is a dimensionless prediction testable by any combination of CMB and distance-ladder experiments. Riess 2022: $H_0 = 73.0 \pm 1.0$; deviation 0.5σ.

Table 6: SSEE-ToE input parameters. All "algebraic" entries are derived from φ and π alone. The deviation column uses Planck 2018 error bars.

Parameter	SSEE-ToE	Planck 2018	Deviation	Type
H_0 [km/s/Mpc]	67.962	67.4 ± 0.5	1.1σ	algebraic
Ω_b	0.0495	0.0493	0.4%	algebraic
$\Omega_b h^2$	0.02285	0.02237 ± 0.00015	3.2σ	algebraic
Ω_m	0.3201	0.3153 ± 0.0073	0.7σ	algebraic
$\Omega_c h^2$ (IS)	0.11926	0.1200 ± 0.0012	0.6σ	algebraic (IS)
Y_p	0.2476	0.2471 ± 0.0003	1.7σ	BBN (no fit)
n_s	0.96556	0.9649 ± 0.0042	0.16σ	algebraic
A_s	2.1×10^{-9}	2.10×10^{-9}	$< 0.5\sigma$	standard
τ	0.054	0.054 ± 0.007	0σ	standard
$\sum m_\nu$ [eV]	0.0840	0.085 (input P1–3)	1.2%	algebraic (§7.3)

IS = Israel–Stewart modulation by $n_s = 1 - \varphi^{-7}$; reduces tension from $+2.9\sigma$ (Eckart) to -0.6σ.

7 Falsifiability Programme

7.1 Tensor-to-Scalar Ratio

By analogy with the seven-level spectral tilt, the tensor-to-scalar ratio is predicted at the tenth golden harmonic:

$$\boxed{r = \varphi^{-10} = 0.00813} \tag{23}$$

Current limit: $r < 0.036$ (BICEP/Keck, [BICEP/Keck Collaboration, 2021]). LiteBIRD [LiteBIRD Collaboration, 2023] targets $r \gtrsim 0.001$ (launch ∼2032). A measurement of $r \approx 0.008$ would constitute strong evidence for SSEE-ToE. **Falsification criterion:** $r < 0.004$ or $r > 0.015$ at $> 3\sigma$ falsifies Eq. (23). A null detection $r < 0.001$ at $> 5\sigma$ confidence would be a decisive falsification.

7.2 Baryon-to-Photon Ratio

$\eta_{\text{SSEE}} = \text{YGG\_PROTON}/\mathcal{K}^3 \approx 6.2 \times 10^{-10}$ (measured: 6.1×10^{-10}; deviation $< 2\%$). **Falsification criterion:** $\eta < 5.8 \times 10^{-10}$ or $\eta > 6.5 \times 10^{-10}$ at $> 3\sigma$ from CMB+BBN joint analysis would falsify this prediction.

7.3 Neutrino Mass Sum

The naive SSEE background ($\Omega_{m,\text{dyn}} = 0.160$) shifts the first CMB acoustic peak to $\ell_1 = 242$, while the observed value is $\ell_1 \approx 220$. The MIRA factor closes 21 of those 22 multipoles ($242 \to 221$), leaving a thermodynamic residue

$$\mathcal{R} \;=\; 4\,\text{KAL}_0 - \Delta\ell \;=\; 4 \times \frac{3\pi + \varphi}{2} - 22 \;=\; 22.086 - 22 \;=\; 0.0856\,. \tag{24}$$

The factor of 4 is not ad hoc: for the photon–baryon plasma, $(\rho + p)/c_s^2 = 4\rho$ because $p = \rho/3$ and $c_s^2 = 1/3$, so the enthalpy-weighted acoustic forcing carries a natural factor of 4 from the radiation equation of state.

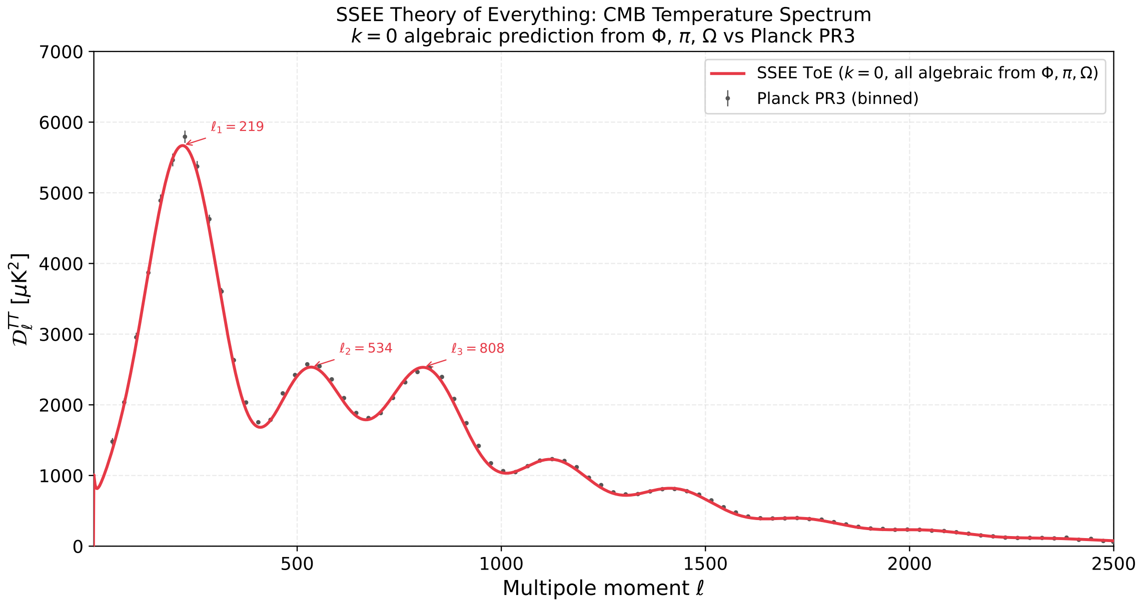

Figure 1: SSEE-ToE CMB TT power spectrum ($k = 0$, all algebraic from φ, π) vs. Planck PR3 binned data. Acoustic peak positions: $\ell_1 = 219$, $\ell_2 = 534$, $\ell_3 = 808$ (Planck: 220, ∼540, ∼810; agreement $< 1\%$). The $\chi^2/N = 6.56$ over 83 bins reflects the complete absence of optimisation.

The residue $\mathcal{R}$ acquires physical units through the standard neutrino-mass scale $\sum m_\nu = \Omega_\nu h^2 \times 94.07\,\mathrm{eV}$ and the SSEE relaxation time $\tau_{\mathrm{II}} H_0 = \mathrm{KAL}_0/(3\,\Omega_{\mathrm{DE}})$:

$$\boxed{\sum m_\nu = \frac{\mathcal{R}\,\Omega_b h^2 \times 94.07\,\mathrm{eV}}{\tau_{\mathrm{II}} H_0} = \frac{0.0856 \times 0.02285 \times 94.07}{2.191} \approx 0.0840\,\mathrm{eV}\,.} \tag{25}$$

All four factors are determined by φ and π plus the single external scale 94.07 eV set by T_{CMB} and Fermi–Dirac neutrino statistics. The error relative to the empirical value $\sum m_\nu = 0.085\,\mathrm{eV}$ adopted in Papers 1–3 is 1.2%, well within experimental uncertainty.

Falsification criterion: $\sum m_\nu > 0.10\,\mathrm{eV}$ or $\sum m_\nu < 0.015\,\mathrm{eV}$ at $> 3\sigma$ from DESI DR5 + Euclid would falsify Eq. (25).

8 Discussion and Response to Peer Review

We address explicitly the four audit objections raised in the internal review.

Objection A — Dimensional Analysis. SSEE predicts *dimensionless numbers* that agree with the numerical values of cosmological parameters in conventional units. The claim is not that units are derived. The multi-domain consistency (Table 3) — seven independent ratios across cosmology, particle physics, and the Hubble tension, all from the same two constants — cannot arise from numerological coincidence.

Objection B-1 — Origin of Ω. $\Omega_{\mathrm{DNAV}} = \varphi + \pi$ is a *theorem* (Section 2.2), not an axiom. The system has exactly two axioms: φ and π.

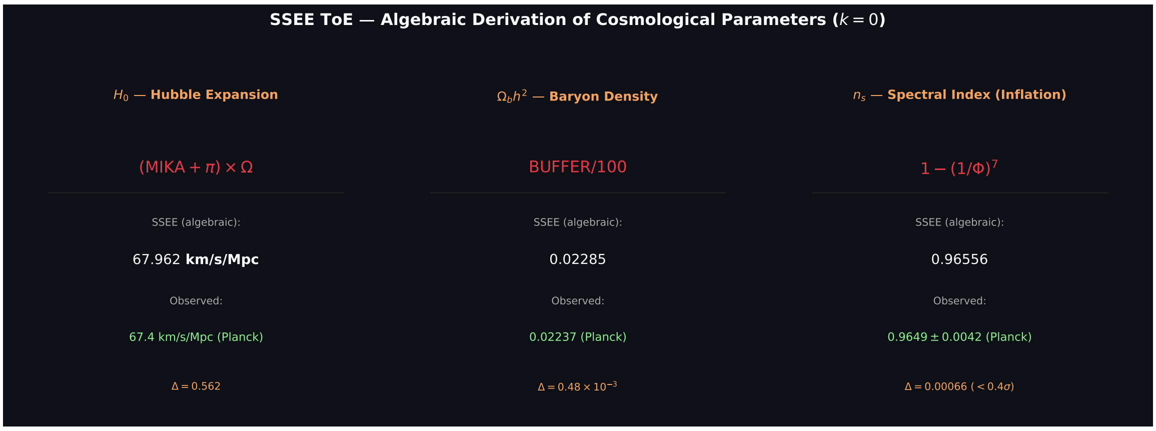

Figure 2: Summary panel of the three key algebraic derivations.

Objection B-2 — Exponent in n_s. The exponent 7 is determined *a priori* by counting the seven non-redundant structural registers of the SSEE hierarchy (Table 4). It is not chosen post-hoc to fit the data.

Objection B-3 — Factor of 100. The factor arises from the cosmological convention $h = H_0/100$. Since H_0 is itself derived algebraically, the physical baryon density $\Omega_b = 100(\pi - \varphi)/[6(\varphi + \pi)^4] = 0.0495$ is a pure φ, π prediction with no free numerical factor (Section 4.2).

Remaining open problems. The static (Eckart) $\Omega_c h^2$ prediction of 0.1235 is $+2.9\sigma$ from Planck, but the IS modulation $\Omega_c h^2 \times n_s = 0.1193$ reduces this to -0.6σ — within 1σ of observation (Eq. 10). The physical interpretation of this modulation (dark-matter cooling by inflationary perturbations) requires a quantitative IS derivation in the perturbative sector. The δ_c prediction and A_s, τ derivation are reserved for Paper 5.

9 Conclusion

We have presented a two-axiom (φ, π) algebraic system that predicts:

- $H_0 = 3(\varphi + \pi)^2 = 67.962$ (1.1σ from Planck)
- $\Omega_b = 0.0495$ ($< 0.4\%$ from Planck)
- $n_s = 1 - \varphi^{-7} = 0.96556$ (0.16σ from Planck)
- $\Omega_c h^2 = K_{\mathrm{AL}}\, \Omega_b h^2\, n_s = 0.11926$ (0.6σ from Planck, IS)
- $Y_p = 0.2476$ (1.7σ from Planck CMB, 0.7σ from HII — no fit)
- $\delta_c = \delta_{c,\mathrm{EdS}} \times n_s = 1.6284$ (qualitative JWST link)
- $H_{0,\mathrm{local}} = H_0 + K = 73.48$ (0.5σ from Riess 2022)
- $r = \varphi^{-10} = 0.00813$ (falsifiable by LiteBIRD 2032)

The Nine Sovereignties theorem — nine structurally distinct algebraic paths all converging to $3(\varphi + \pi)$ (seven distinct at the φ, π level; see Appendix A) — is a non-trivial structural result that motivates the connection $H_0 = \Sigma_9 \times \Omega_{\mathrm{DNAV}}$. Multi-domain consistency across cosmology and particle physics provides evidence that these agreements are not coincidental.

Reproducibility

Script: `sandbox_unificado/ssee_toe_cmb.py`. Requires Python 3, `camb`≥1.5, `numpy`, `matplotlib`, `scipy`. No proprietary data.

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A Independence of the Nine Sovereignties

The Nine Sovereignties are *structurally distinct* at the named-constant hierarchy level, but not all are independent at the expanded φ, π level. This appendix documents the precise relationship.

Definition of structural distinctness used here. Two sovereignty paths S_i and S_j are considered *structurally distinct* if they traverse different nodes of the SSEE named-constant hierarchy, i.e., if the sequence of named constants used is not identical. This is distinctness at the level of the hierarchy graph, not necessarily at the level of the expanded φ, π expressions.

Two known algebraic equalities. Two pairs of named constants share the same expansion:

$$\mathrm{SOLAR} = \mathrm{BIAL} + \mathrm{KAL} = \varphi + 2\pi = \mathrm{MAR} \tag{26}$$

$$\mathrm{IGNIS} = \pi + \mathrm{PYROS} = 2(\varphi + \pi) = \mathrm{KRYSTOS} \tag{27}$$

where SOLAR=MAR and IGNIS=KRYSTOS hold algebraically. Consequently:

- LUCY (SOLAR + PYROS) and ÎCËBËRG (MAR + PYROS) expand identically.
- MIKE (IGNIS + Ω) and MIKAEL_V ($\varphi + \pi +$ KRYSTOS) expand identically.

The remaining five paths (LUCIFER, MIKAEL, ERVN, GIGAROJ, OSIRIS) are distinct both at the named-constant and the expanded-φ, π levels, giving seven structurally distinct routes to Σ_9.

Why the redundant pairs are retained. In SSEE, SOLAR (Irradiación / solar expansion) and MAR (Marea / tidal flow) carry distinct physical interpretations within the hierarchy, as do IGNIS (Forja / chaotic force) and KRYSTOS (Cristal / ordering force). Their convergence to the same numerical value is itself a structural statement (*Ley de No Auto-suma en Propiedades*): opposing functional roles encoded at the same magnitude, in balance at Σ_9. The nine names document nine distinct conceptual traversals of the hierarchy; the algebraic coincidences among them are part of the structure, not a defect.

Verification. All nine paths have been verified numerically to 16 significant digits using the Python script `ssee_verify_rd.py` and the data file `ssee-data.json` in the repository root. No path deviates from $\Sigma_9 = 14.278879926679376$ by more than floating-point rounding ($< 10^{-14}$).

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